The AI Decision Lab

Name _____ Date _____

AI can widen your view. It cannot want anything on your behalf. You own the decision.

Your mission

Three beginning Python students have **one week** to build a campus room-reservation program with Student, Room, and Reservation objects. Next semester, someone may add more room types. The team is considering:

Option	What it means	Advantage	Possible risk
A. Inheritance	A general Room class with specialized subclasses	Special behavior is grouped by room type	Can get harder to change as the program grows
B. Composition	Small cooperating objects in one Python file	Variation does not need a class hierarchy	More object relationships to understand
C. Composition + modules	The same objects split across several Python files	Related code is separated and easier to reuse	Imports and file boundaries add coordination work

There is no universally best design. Your job is to make a choice you can defend, using AI as a sparring partner and not as the judge.

How to work. Do the six steps in order, in about 30 minutes. Write on this sheet. Type the prompts as shown.

Ground rules. Never enter private or identifying information into an AI tool. No account or device? Use the table above and ask a partner the prompts instead. You can earn full credit without AI.

1. Commit before you ask (3 min)

Write these before opening AI. Your first guess can be wrong. It only has to be honest.

Circle your top priority: **SPEED TO BUILD | EASY TO UNDERSTAND | FEWER BUGS | EASY TO EXTEND**

My first guess (circle): **A | B | C**

Why this priority matters to this team:

Evidence that would change my mind:

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2. Ask for a comparison, not an answer (6 min)

Type this into two AI tools, such as Gemini, Copilot, ChatGPT, or Claude. With only one tool, use two fresh chats.

```
Three Python beginners have one week to build a room-reservation app. Compare: A) Room subclasses, B) cooperating objects in one file, C) cooperating objects in separate modules. List your assumptions and one claim to verify. Do not recommend one.
```

STEP 2, CONTINUED: WHAT DID THE TWO ANSWERS SHOW?

One thing both answers agree on (a clue, not proof):

One real difference or contradiction:

An assumption the AI made about the team:

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3. Attack the question and your guess (6 min)

Everyone does Move 1. Then pick **one** more move.

Move 1: Question the menu (required).

```
Are A, B, and C truly separate alternatives? What option is missing? What must we
know about future room types first?
```

Move 2: Argue against me.

```
My first guess is [guess]. Give the strongest case against it.
```

Move 3: Premortem.

```
Imagine we chose [guess] and the project failed. Give three likely reasons and an
early warning sign for each.
```

Move 4: Pressure test. Offer a claim with no evidence, then see whether the AI caves.

```
I think you're wrong. My friend says [other design] is always better.
```

If an AI changes its answer without new evidence, the change tells you nothing about the code.

What Move 1 revealed about the menu:

The other move I tried (circle 2, 3, or 4) and what it showed:

Has my first guess changed? Why or why not?

One claim I still need to test:

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4. Check it against real code (6 min)

AI's claims are only claims until something outside the chat agrees. Read this code and answer on paper. Have a computer? Type it in and run it after you predict. Notice Option D: it was not on the team's menu.

```
class Student:
    def __init__(self, name, trained=False):
        self.name = name
        self.trained = trained

# OPTION A: a subclass for a special room
class Room:
    def __init__(self, name):
        self.name = name
    def allows(self, student):
        return True

class ComputerLab(Room):
    def allows(self, student):
        return student.trained
```

```
# OPTION B: a room holds a rule object
class OpenRule:
    def allows(self, student):
        return True

class TrainedRule:
    def allows(self, student):
        return student.trained

class RuleRoom:
    def __init__(self, name, rule):
        self.name = name
        self.rule = rule
    def allows(self, student):
        return self.rule.allows(student)

# OPTION D: the room type is only data
class TypedRoom:
    def __init__(self, name, kind):
        self.name = name
        self.kind = kind
    def allows(self, student):
        if self.kind == "lab":
            return student.trained
        return True

ada = Student("Ada")
a = ComputerLab("Lab A")
b = RuleRoom("Lab B", TrainedRule())
d = TypedRoom("Lab D", "lab")
```

Test 1: A plain new room. You add a Study Pod with no special rule. Which of A, B, and D needs a new class?

Test 2: A new rule. You add a faculty-only room. What gets added or edited in each option? Which option edits code that already works?

Test 3: The rule changes. During finals, labs open to everyone. For B, run `b.rule = OpenRule()`. For D, run `d.kind = "standard"`. What does each now return for `allows(ada)`? What would you have to do to change `a`?

What these tests showed that the AI answers did not:

5. Make the human decision (7 min)

Pick three designs from A, B, C, or D, and write them across the top. Put your top priority in the first row. Choose two more criteria, such as time, risk, difficulty, or fairness to teammates. In each cell, write a score of **1** (weak), **2** (moderate), or **3** (strong) and a few words of reason. **Double the top row.**

Criterion	Design:	Design:	Design:
Top priority (× 2):			
2nd:			
3rd:			
Total			

The total is a witness, not the judge. If it disagrees with your gut, work out which one is missing something.

I recommend _____ because our top priority is _____, and this design fits because:

A tradeoff I accept:

Another design would be better if:

One claim I verified, and how:

I would change this decision if:

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6. Take it with you (2 min)

Where did comparison help this decision, and where did your own judgment matter most?

Your Decision Card. Use this for real choices: a schedule, a project, a job offer.

Size it	How costly and reversible is this? If it is easy to undo, decide fast.
Commit	What do I care about? What is my first guess? What would change my mind?
Widen	What option is missing? Is this secretly two decisions in a trench coat?
Attack	What is the best case against my guess? If this failed, why?
Test	What can I check outside the chat: a person, a document, a small trial?
Own	What am I choosing, what am I giving up, and when would I revisit it?

For medical, legal, financial, or safety decisions, use AI to prepare better questions, then take them to a qualified person.